

process begins are the same — taking a back-up! Let me make it clear. You must frequently take a back-up of your valuable data to keep it safe (i.e., no program should be allowed to work on the back-up medium). If you take this precaution, no crash can break your heart. But what if you haven't taken a back-up? What if you want your true system back instead of a duplicate? In that case, let us try some tricks. But these can sometimes kill your data, which is available in the crashed file system even though it doesn't seem to be accessible. That is why you need to take a back-up before trying out these tricks. The question is, how do you take back-ups from a crashed file system?

Sometimes, for example, only the / (root) partition becomes damaged and the /home partition, which holds your personal data, is safe. In such cases, you will be unable to boot your native OS but will still be able to access the data using a 'recovery disk'. I will discuss how to handle such situations without data loss later. Meanwhile, it is recommended to take a back-up for safety.

Find out which trick works best

a) Simple copying: This is performed when you can access the disk drives from a recovery disk even though you are unable to boot the native OS.

1. Simply boot from a live CD without installing the OS or boot another OS in the hard disk.
2. Use the native file browser to open the available drives/partitions of the hard disk (you can find them inside 'Computer' in an Ubuntu live CD).
3. Connect a USB mass storage device (or something similar) and copy the data. You can also connect it with a network and take the back-up. Remember, this is not about 'sharing files' but 'copying files'.

b) Using software tools: You can use programs like Déjà Dup to take back-ups. Lost files can be recovered with PhotoRec or similar tools, which we will explore in the next section.

c) Taking the file system image: A file system image is like a CD image, which keeps each byte of the disk. Truly being an 'image', it can also be treated as your damaged file system. When you crash your 'already damaged' file system via the recovery process, you can take the copies of your fs image (which you ideally will have taken before your experiments), send it to some experts and challenge them to recover it. The process is simple:

1. Simply boot from a live CD without installing the OS or boot another OS in the hard disk.
2. Note the names of partitions like /dev/sda1, whose images are to be taken. Normally, /dev/sda1, /dev/sda2, etc., are the names of hard disk partitions. Additional USB storage media are provided with /dev/sdb1, /dev/sdb2, etc. 'sdb' will be your first USB device and 'sdc' the second. Names without numbers point to the whole device (e.g., /dev/sda means the entire hard disk) and



Figure 1: Disk utility shows the details about hard disk partitions

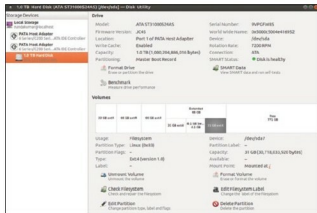


Figure 2: Home screen of Déjà Dup

the numbers point to the partitions. You can use disk utilities to find these names.

- CAUTION:** If you find and follow a wrong partition number, it'll cause data loss.
3. Unmount the partitions whose recovery is to be done (if you have mounted any).
 4. Open a terminal and give commands in the following format:

```
sudo mount DEVICE MOUNT_POINT
sudo genisoimage -o OUTPUT-FILE MOUNT_POINT
sudo umount DEVICE
```

Example:

```
sudo mount /dev/sda2 /mnt
sudo genisoimage -o /media/MyUSBDevice/sda2.iso /mnt
sudo umount /dev/sda2
```

In the above example, /dev/sda2 is assumed as the /home partition of a crashed GNU/Linux distro and /media/MyUSBDevice is taken as the back-up medium (normally, USB storage devices are mounted into /media with their labels; if you don't find them mounted, use the mount command or disk utilities to mount them).